VED LOYA

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PROFESSIONAL SUMMARY

Computer Engineering student with hands-on experience in Python, machine learning, and data preprocessing. Currently learning and building ML projects, with an interest in model development and backend development using FastAPI. Seeking opportunities to apply my technical skills, gain real-world experience, and keep growing in machine learning.

EDUCATION

B.E. in Computer Engineering 2024 – 2028

Ramdeobaba University, Nagpur, Maharashtra

- CGPA: 7.77 (currently in 5th semester)

HSC 2024

Deogiri College

- Percentage: 82.33%

PROJECTS

Software Defect Risk Engine Aug 2026 – Sep 2026

Tech Stack: Python, XGBoost, SHAP, FastAPI, SQLAlchemy, Radon, Pandas | github.com/vedloya11/software-defect-risk-engine

- Built an ML system that predicts the defect risk (Low / Medium / High) of Python source files, trained on the NASA KC1 benchmark dataset (2,109 modules, about 15% defective).
- Created extra features such as operand and operator density, compared Logistic Regression, Random Forest, and XGBoost, and tuned XGBoost with 5-fold cross-validation and class-imbalance handling.
- Achieved **0.82 ROC-AUC** and **84% recall** on a held-out test set, focusing on recall so most defect-prone code gets flagged.
- Used SHAP to explain each prediction by showing which metrics, such as cyclomatic complexity or branch count, raised or lowered the risk score.
- Wrote a code analyzer using Python's `ast` module and Radon to extract lines of code, complexity, and Halstead metrics from uploaded files.
- Built a FastAPI backend with a web dashboard for uploading code, viewing risk results, and saving scan history in a database using SQLAlchemy.

Used Car Price Predictor Jul 2026

Tech Stack: Python, XGBoost, scikit-learn, Pandas, FastAPI, HTML/CSS/JavaScript | github.com/vedloya11/Car_Price_Predict

- Built a model that predicts the fair market price of a used car from 2,059 Indian listings with 20 features such as brand, fuel type, kilometers driven, and engine details.
- Cleaned the data by filling missing values and extracting numbers from text columns like Engine, Max Power, and Max Torque; created features such as Car Age and encoded 8 categorical columns.
- Trained an XGBoost Regressor with an 80/20 train-test split, reaching an **R^2 of 0.89** (mean absolute error of about Rs. 2.66 lakh).
- Added a K-Nearest Neighbors recommender that suggests the 5 real listings closest to the predicted price.
- Deployed the model with FastAPI and a simple web page where users enter car details and instantly see the predicted price.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, SHAP, Feature Engineering, Model Evaluation
Data: Pandas, NumPy, Matplotlib, Seaborn
Backend: FastAPI, Pydantic, SQLAlchemy
Databases: PostgreSQL, Oracle, MySQL
Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter, Pytest

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Treasurer, ACM Student Chapter, RBU**
- **Joint Venue Manager, Compusys (CSE Branch Committee)**
- **NSS Volunteer**, National Service Scheme